

Code-quality and frontend QA report

Repository: kortix-ai/suna

Reviewed: 2026-08-28

Original head: 5c609e689e03f2a93468e32257ac537e7e1f1517

Fix commit: 7cc805c37578d66c50ef9ebed1c76c58ed45eb64

Current head: 1f86186e6d0c9c526960e9b18834e65395e7d1fc

DECISION: DO NOT MERGE YET

The confirmed code gaps are fixed. Queue persistence and refresh visibility pass in Chromium. The cloud runtime ended with runtime_gone before the second prompt completed. New-head CI is still running.

Executive result

- JAY-720 now retries history from the last successful cursor.
- JAY-721 now bounds the first never-settling /start request.
- The lifecycle queue now sends one prompt per session.
- Empty zero-token assistant rows no longer paint blank turns.
- Older empty queue snapshots cannot erase a newer queue row in the SDK observation path.
- The branch is synchronized with main. GitHub reports mergeable: true.

Confidence

Code-level confidence is high for the fixed regressions. User-flow confidence is medium. Ship confidence remains blocked by one stable two-prompt completion and new-head CI.

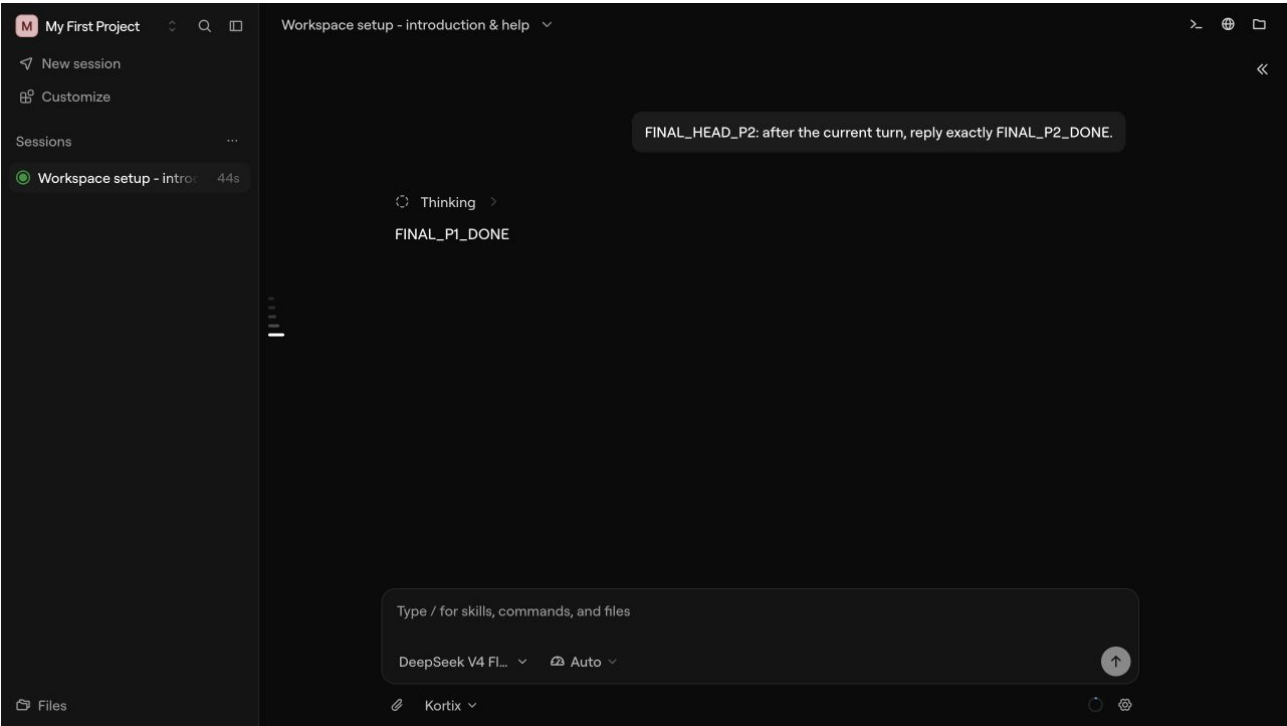
Original-head browser failure

P1 requested a 30-second wait. P2 was submitted while P1 was active. Both lifecycle rows were marked delivered, but the first answer appeared below P2. The requested P2 result never appeared.

Prompt	Started UTC	Ended UTC	Duration
P1	17:43:26.534	17:43:59.945	33.411 s
P2	17:43:38.791	17:44:01.254	22.463 s

Confirmed integrity failure

The turns overlapped for about 21.154 seconds. The UI attributed FINAL_P1_DONE to P2. No assistant message contained FINAL_P2_DONE.



Original PR head: the P1 result appears below the P2 prompt.

Fixes implemented

JAY-720 - history cursor

- Each successful page hydrates messages and commits its cursor.
- A failure after cursor 5 now retries with before=cursor-5.

JAY-721 - first unresolved start

- useSessionStartGiveUp owns the deadline and verdict.
- The first unresolved request arms the deadline.
- Later data or errors clear the deadline and verdict.

Durable queue serialization

- An active turn refuses admission with reason turn_active.
- One drain sends only the head prompt for each session.
- Sibling prompts return to the queue with bounded backoff.
- The engine no longer starts same-session deliveries concurrently.

Related correctness fixes

- Parentless zero-part, zero-token assistants are skipped only when they have no error.
- Queue projection and query cache updates share applyInboxObservation.
- splitTimingDurations removes wall-clock subtraction from timing assertions.

Post-main-sync regression gates

Full SDK suite: pass. SDK typecheck: exit 0. Focused API queue and timing tests: 38 pass, 0 fail.

Post-fix browser evidence

P1 requested a 20-second wait and FIRST_OK_0828. P2 requested SECOND_OK_0828 while P1 was active. Both POST /prompts calls returned HTTP 202. GET /prompts retained P2 with reason turn_active and attempts 0.

P2 remains queued while P1 owns the active turn.

The screenshot displays the OpenAI ChatGPT web interface. On the left is a sidebar with a 'FIRE' header and navigation options: 'New session', 'Customize', 'Apps', 'Sessions', 'Today', 'Yesterday', 'Review', and 'Files'. The 'Sessions' list includes 'Genuine queueing for second message' (6m), 'New session' (8m), 'New session' (30m), and several sessions from 'Yesterday' such as 'New session' (2h), 'Friendly greeting' (2h), 'QUAGGATOKEN single-' (2h), 'ZEBRAMARKER response' (2h), 'Daily AI Workforce' (2h), 'Friendly greeting' (2h), 'Exact PONG reply' (3h), 'Verify AI Workforce' (3h), and 'Single-word PONG response' (3h). The 'Review' section has a yellow badge with the number '4'. The main chat area shows a conversation with the session 'Genuine queueing for second message'. The first message is a system prompt: 'QUEUE_FIX_PL_0828: Run the terminal sleep command for 20 seconds. After it completes, reply with exactly FIRST_OK_0828 and nothing else.' The user's response is '> Ran command sleep 20'. The assistant's reply is 'FIRST_OK_0828' followed by a thinking state 'Gathering thoughts...'. The second system prompt is 'QUEUE_FIX_P2_0828: Reply with exactly SECOND_OK_0828 and nothing else.' The input field at the bottom contains the text 'Type @ to mention files and agents' and shows 'GPT-5.6 Sol (Ch...' and 'Auto' as model and tone options. The bottom status bar shows 'Kortix' and system icons.

The queued P2 row remains visible after a browser refresh.

Browser verdict

Verified

P2 stayed in the durable queue. The UI displayed its queued state. A refresh preserved the row. No same-session overlap occurred during the observed window.

Not verified

The runtime ended with `runtime_gone`. P1 was requeued as redelivered after `runtime_gone`. P2 remained waiting. The run did not prove automatic P2 completion or final parent-to-answer attribution.

Automated verification

Surface	Result
Focused SDK regression files	71 pass, 0 fail
Mounted start give-up hook	2 pass, 0 fail
Full SDK suite after main sync	Pass
SDK typecheck after main sync	Exit 0
SDK packed install smoke	Exit 0
Focused API after main sync	38 pass, 0 fail
Full API before main sync	8,748 pass, 79 skip, 0 fail
Full web before main sync	8,648 pass, 0 fail
Standalone CLI before main sync	1,243 pass, 0 fail
Changed-file ESLint	0 errors, 32 warnings
git diff --check	Clean

The root packages-only run failed in its concurrent CLI sub-lane without useful output. The same CLI suite passed standalone. Treat the root package lane as unresolved until CI completes.

Linear Phase 0 verification

Issue	Assessment
JAY-717	Unit-covered; real provider retry not induced
JAY-718	Long-turn Stop verified; provider retry not induced
JAY-719	Keepalive excluded from process evidence
JAY-720	Fixed and verified
JAY-721	Fixed and verified
JAY-722	Active provisioning browser verified
JAY-723	Unit-covered; real poison sequence not induced
JAY-724	Boot-phase header browser verified
JAY-725	Cache-key covered; custom directory browser run absent
JAY-726	Empty assistant suppression fixed by tests
JAY-291	Legacy queue serialized; protocol v2 remains

Status discipline

All PR-related issues remain In Progress. Repository policy permits Done only after merge, dev deployment, and dev verification.

New tracked gaps

JAY-728 - queue snapshot freshness clock

POST, GET, bundle, and stream observations need one server-defined order or revision.

JAY-729 - failed-session redirect escape

A failed session needs a deterministic browser test, a visible escape action, and no redirect loop.

Marko critical-issue mapping

Items	Theme	Tracking / assessment
1-3	Mid-turn death and silent death	JAY-430, JAY-566, JAY-723; not closed
4	140 MB history and OOM	JAY-686, JAY-689, JAY-351; not closed
5	UI slower than terminal	JAY-697; not closed
6	Agent and stream architecture	Protocol phases 1-4; Phase 0 only
7-10	Queue and reconnection	JAY-291, JAY-572, JAY-726; improved
11	Warm prior-session regression	JAY-596; not closed
12	Slow resume	JAY-697; not closed
13	Redirect softlock	JAY-729; new urgent ticket
14-15	Stop/restart and bricked session	JAY-430, JAY-566, JAY-723; not closed
16	Unexplained terminal incident	Needs incident-specific evidence

Scope conclusion

PR #6957 is Phase 0 symptom relief. It does not complete the Session State Protocol architecture or close all 16 incidents.

Thermo-nuclear code-quality review

Correct improvements

- The new start give-up hook removes timing state from the 1,500-line use-session hook.
- Queue observations pass through one helper per observation path.
- The false stream-attached polling mode is removed.
- The terminal card gate uses the SDK SessionStartStage type.
- Reconciler comments now match the actual four subsystem reads.

Remaining structural debt

Production file	Lines
apps/web session-chat.tsx	5,433
daemon main.ts	3,234
SDK sync-store.ts	2,723
lifecycle engine.ts	2,244
SDK use-session.ts	1,521
session route page.tsx	1,417

Review boundaries

No PR file crossed from below 1,000 lines to above 1,000 lines. Source-text assertions remain weaker than mounted or black-box behavior tests. The unrelated globals.css text-blur removal should move to a separate PR.

Final shipping gate

Do not merge PR #6957 until every item below passes.

- GitHub checks pass at head 1f86186e6d0c9c526960e9b18834e65395e7d1fc.
- A stable real sandbox completes P1 and P2 automatically.
- The DOM contains both exact assistant outputs under the correct prompts.
- Lifecycle timestamps prove no same-session overlap.
- Refresh during P1 preserves P2 and later drains it.
- JAY-728 and JAY-729 remain tracked until their acceptance tests pass.
- After merge, dev deploys the merged SHA and the same browser journey passes on dev.

Current recommendation

Keep the PR open and unmerged. The confirmed code defects are fixed. Wait for CI and rerun the two-prompt browser journey on a stable runtime.

No credentials are stored in this PDF. No PR merge occurred.